

Patient Flow & Bed Occupancy Analytics

Hospital Operations Data Platform Requirements

From: Dr. Sample Doctor - Chief Operations Officer (COO), Midwest Health Alliance

To: Data Engineering Solutions Partner

Subject: Development of a Real-Time Data Platform for Patient Flow & Bed Occupancy

Business Background

Midwest Health Alliance (MHA) is a network of **7 hospitals** across the Midwest region. We face daily challenges in **patient flow management**, especially during high-demand periods such as flu outbreaks.

We lack a centralized, real-time data system to monitor **bed occupancy**, **patient admission/discharge patterns**, and **department load**.

Business Objectives

- **Monitoring patient admissions** to minimize waiting times.
- **Identify department-level bottlenecks** (e.g., Emergency, Surgery, ICU).
- **Enable gender-based and age-based KPIs** for demographic insights.
- **Automate alerts in case of failures.**

Functional Requirements

Data Sources

- Real-time patient admissions/discharge data from hospital registration systems.
- Daily batch extracts from Electronic Health Records (EHR) systems.
- Department metadata (capacity, staff numbers).

Data Processing & Storage

- Store data in a **medallion architecture** (Bronze → Silver → Gold).
- Handle **schema evolution** when new patient attributes are introduced.
- Implement **Slowly Changing Dimension Type 2 (SCD2)** for patient and department history.
- Implement **star schema** for storing the tables.

Analytics

- Use **Azure Synapse** for analytics queries.
- Build dashboards in **Power BI** with KPIs such as:
 - Current occupancy % by department
 - Gender-based occupancy distribution
 - Average length of stay per patient

Orchestration & Automation

- Use **Azure Data Factory (ADF)** to automate:
 - Daily batch ingestion from EHR
 - Real-time processing triggers
 - Gold-layer refresh for dashboards

Data Quality

Simulate realistic dirty data issues (e.g., missing admission times, duplicate patient IDs, wrong timestamps) and handle them in the Silver layer processing.

Security & Compliance

Role-based access control for different hospital departments.

Deliverables

- Fully functional Azure-based data pipeline.
- Power BI dashboard connected to live Synapse queries.
- Data quality & validation reports.
- Full project documentation (architecture, data models, Git repo).

Success Criteria

- Dashboards refresh often for real-time views.
- All pipelines fully automated via ADF.
- Schema changes do not cause downtime.